

Semester: IV
Subject: DECSO-BCA-402
Duration: 90 mins

Session: 2024-2027
Course: BCA
Max Marks: 20

SECTION A**(05 × 1 = 05 Marks)**

All questions are compulsory.

1. What is a data structure?
2. Define an array.
3. What is a linked list?
4. Define stack and mention one of its applications.
5. What is the difference between linear and non-linear data structures?

SECTION B**(03 × 05 = 15 Marks)**

Attempt any THREE out of the FOUR questions. Each question has two parts (a) and (b), carrying 5 marks each.

Q1.

- (a) Explain the concept of time complexity and space complexity with suitable examples.
- (b) Explain different types of arrays and describe insertion and deletion operations in an array.

Q2.

- (a) What is a linked list? Explain singly linked list with a neat diagram.
- (b) Explain insertion and deletion operations in a singly linked list.

Q3.

- (a) Define stack. Explain its basic operations (PUSH and POP) with an example.
- (b) Explain the applications of stack in expression evaluation and parentheses matching.

Q4.

- (a) Define queue and explain its basic operations with a suitable example.
- (b) Differentiate between linear queue and circular queue. Explain the advantages of a circular queue.

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